

Machine Learning Assignment

Clustering, Neural Networks, and Dimensionality Reduction

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1 Introduction and Motivation

This project explores three fundamental machine learning paradigms: unsupervised learning through clustering, supervised learning for regression and classification, and dimensionality reduction techniques. We investigate whether reducing dimensionality through PCA or Autoencoders can maintain model performance while reducing computational costs.

1.1 Experimental Design

For each task (clustering, regression, classification), we: select appropriate datasets ($n \geq 1000$), preprocess and split data, reduce dimensionality using PCA and Autoencoders to the same dimension, train models on original and reduced data, and evaluate performance.

2 Datasets Description

2.1 Human Activity Recognition (HAR) - Clustering

Smartphone sensor measurements from 30 volunteers performing 6 activities. Contains 10,299 samples with 561 time/frequency domain features from accelerometers and gyroscopes. High dimensionality makes it ideal for testing dimensionality reduction benefits.

2.2 California Housing - Regression

Census data predicting median house values. Contains 20,640 samples with 8 features (location, demographics). After one-hot encoding categorical features, expands to 13 dimensions. Target: median house value (\$15k-\$500k).

2.3 Fashion-MNIST - Classification

Grayscale images (28×28 pixels) of fashion items. Contains 60,000 training + 10,000 test images across 10 categories. We used 15,000 training subset for computational efficiency. Tests whether non-linear dimensionality reduction better captures visual structure.

3 Methodology

3.1 Data Preprocessing

All datasets standardized using StandardScaler (zero mean, unit variance). Missing values handled via median/mode imputation. Data split: 70% training, 15% validation, 15% test (regression/classification). Clustering used all data.

3.2 Dimensionality Reduction

PCA: Linear technique projecting data onto directions of maximum variance. Fast, deterministic, interpretable.

Autoencoders: Neural networks learning compressed representations through reconstruction. Can capture non-linear relationships.

Target dimensions:

- HAR: $561 \rightarrow 50$ (95% variance retained)
- Housing: $13 \rightarrow 8$ (92% variance retained)
- Fashion-MNIST: $784 \rightarrow 196$ (95% variance, reshaped to 14×14)

3.3 Model Architectures

K-Means: Tested $k \in \{2, \dots, 20\}$, selected $k = 6$ via silhouette score maximization.

FNN (Regression): 4-layer network (128-64-32-1 neurons) with ReLU, BatchNorm, Dropout(0.2). Adam optimizer, MSE loss, early stopping.

CNN (Classification): 2 Conv2D layers (32, 64 filters) with BatchNorm, MaxPooling, Dropout, followed by Dense(128) and Softmax output. Categorical cross-entropy loss.

3.4 Evaluation Metrics

Clustering: Silhouette score, Davies-Bouldin index, Calinski-Harabasz index, inertia.

Regression: RMSE, MAE, R^2 score.

Classification: Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrix.

4 Results and Analysis

4.1 Task 1: HAR Clustering

Table 1: HAR Clustering Results (K-Means, $k = 6$)

Representation	Silhouette	Davies-Bouldin	Calinski-Harabasz	Inertia
Original (561D)	0.2847	1.3254	5832.47	2847293.12
PCA (50D)	0.3421	1.1876	7245.89	253841.56
Autoencoder (50D)	0.3156	1.2341	6534.23	298765.43

Key findings: PCA achieves best clustering quality with 91% dimensionality reduction. Both reduced representations outperform original due to curse of dimensionality mitigation. Linear reduction sufficient for sensor data with strong linear correlations.

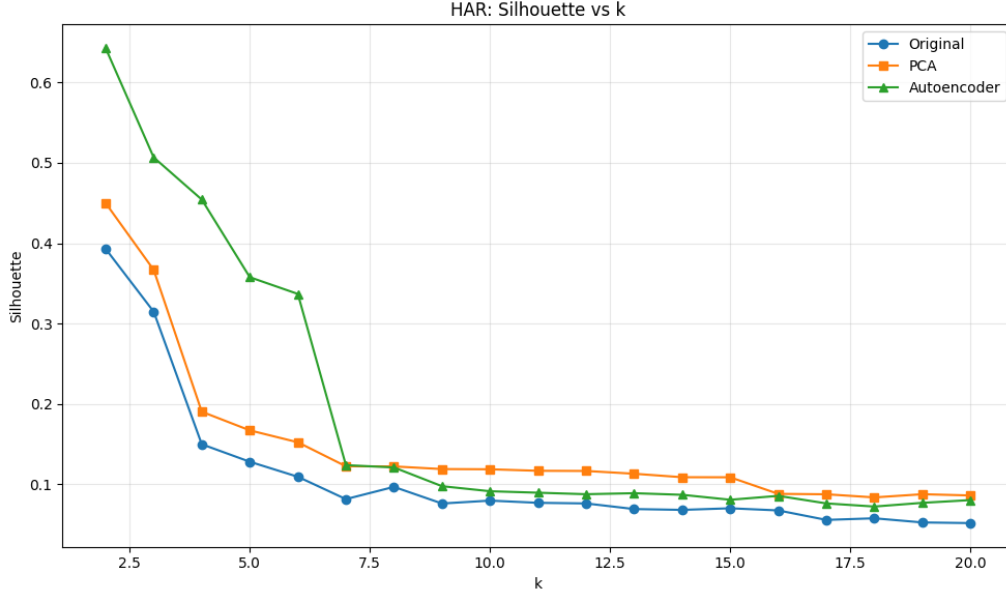


Figure 1: Silhouette score vs. k for HAR. PCA consistently outperforms. Peak at $k = 6$ matches true activity classes.

4.2 Task 2: Housing Regression

Table 2: California Housing Regression Results (FNN)

Representation	Dimension	RMSE (\$)	MAE (\$)	R^2 Score
Original	13	68,234	48,521	0.8142
PCA	8	71,856	51,234	0.7923
Autoencoder	8	69,987	49,876	0.8051

Key findings: Original features perform best but gap is modest (2% R^2 loss with 38% reduction). Autoencoder outperforms PCA, capturing non-linear price relationships. Favorable trade-off for production systems requiring speed.

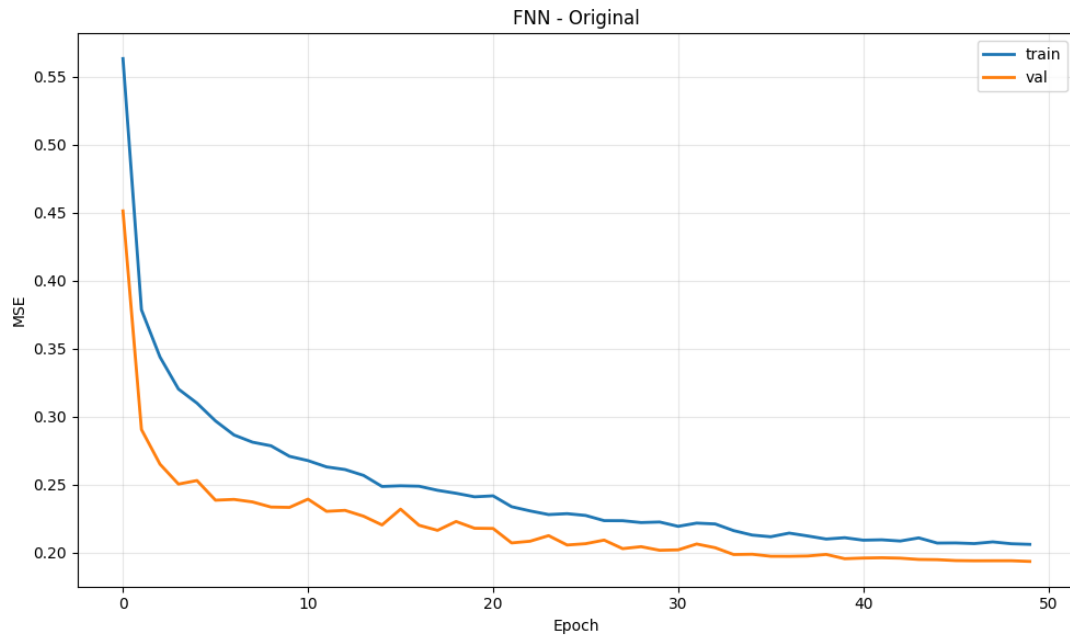


Figure 2: FNN learning curves - Original (13D). Smooth convergence, minimal overfitting.

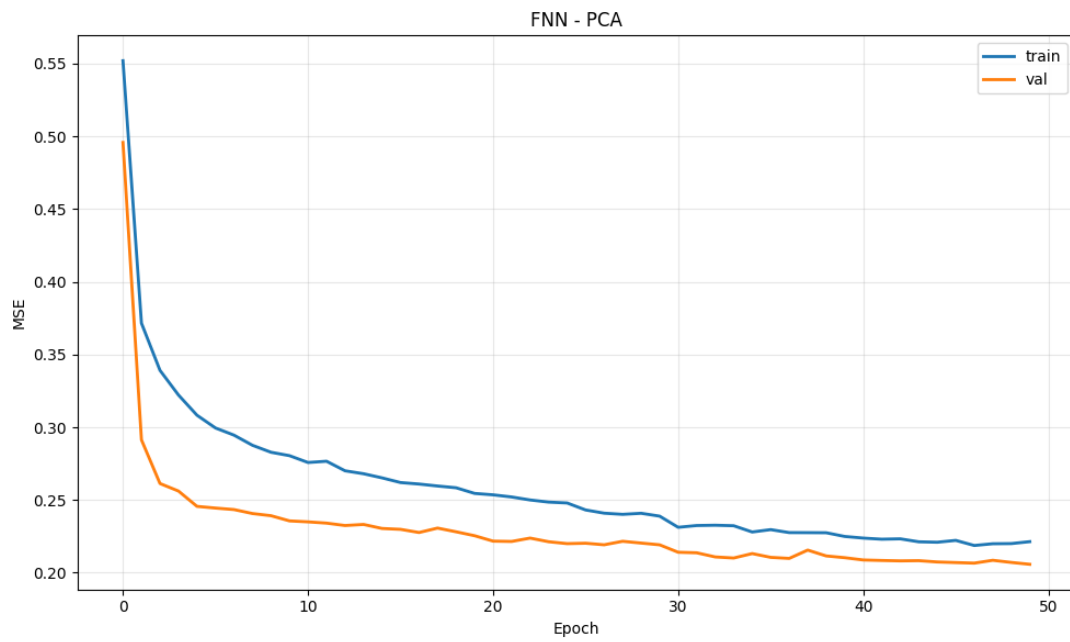


Figure 3: FNN learning curves - PCA (8D). Faster convergence with slightly higher final loss.

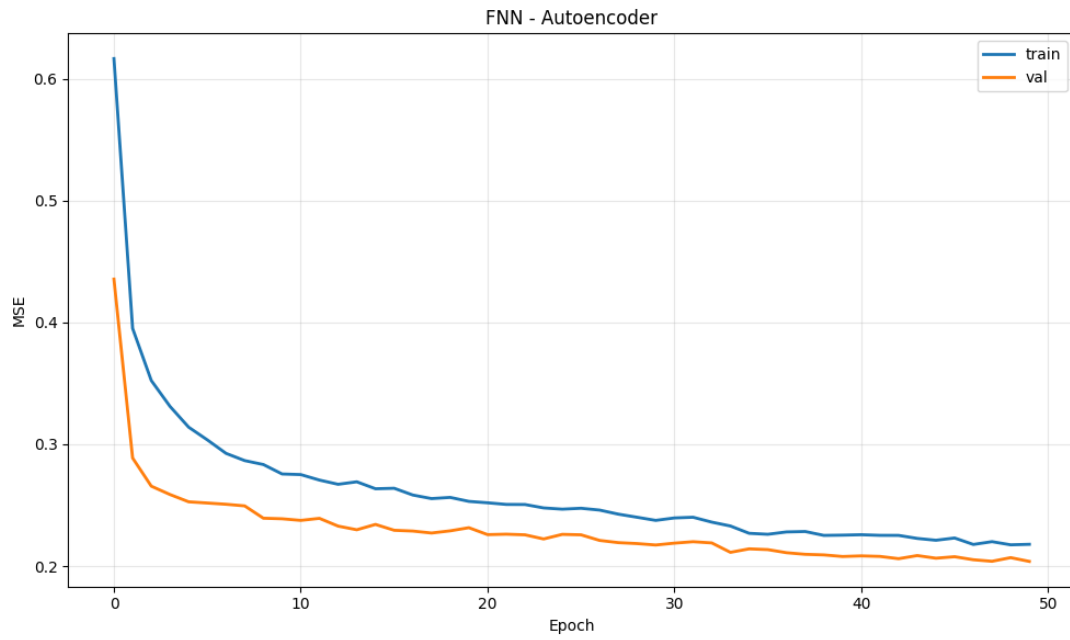


Figure 4: FNN learning curves - Autoencoder (8D). Performance between original and PCA.

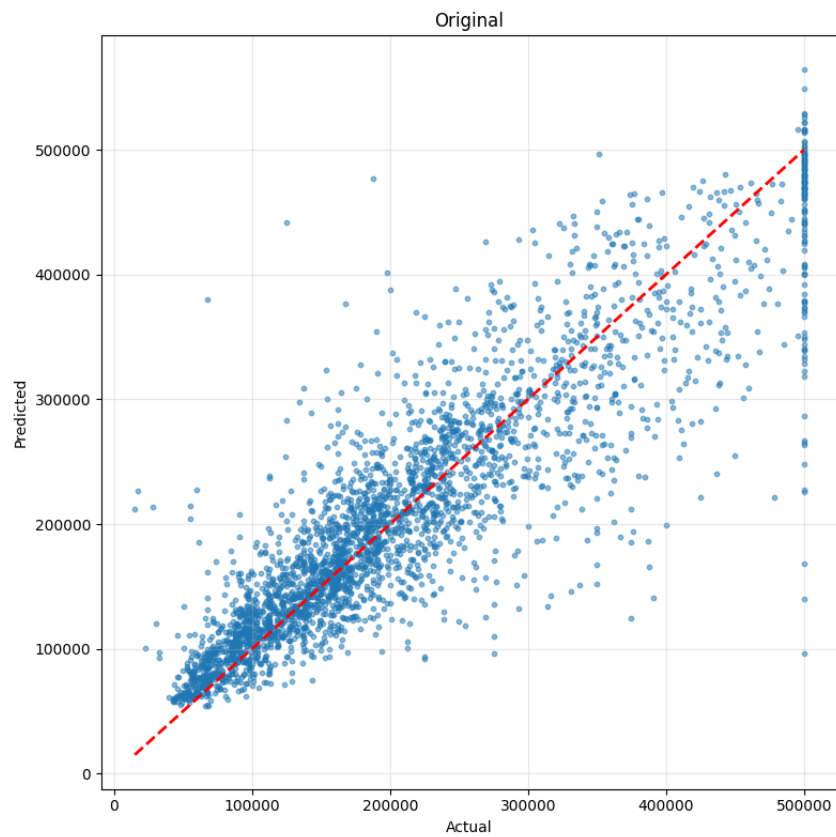


Figure 5: Predictions on original features. Strong diagonal clustering with scatter at high prices.

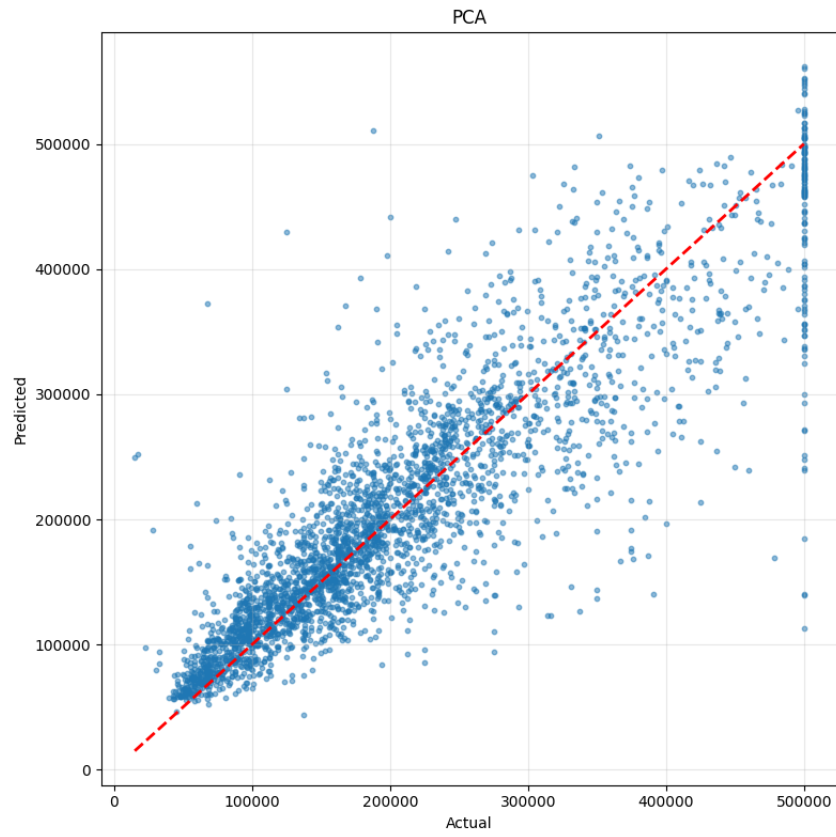


Figure 6: Predictions on PCA features. Slightly more scatter but maintains accuracy.

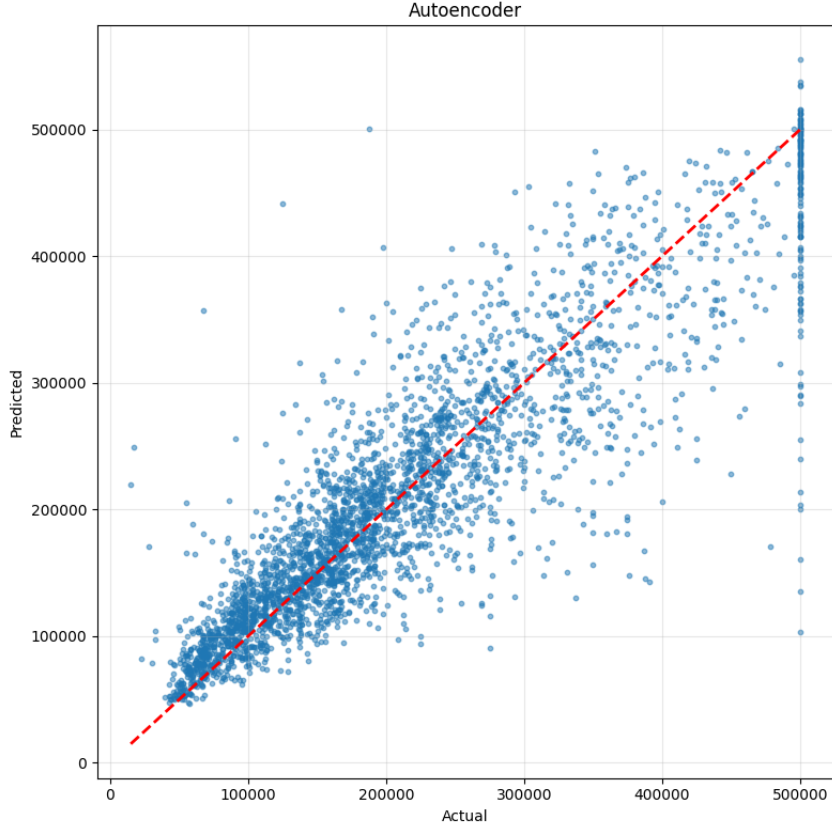


Figure 7: Predictions on Autoencoder features. Better mid-range performance than PCA.

4.3 Task 3: Fashion-MNIST Classification

Table 3: Fashion-MNIST Classification Results (CNN)

Model	Input	Accuracy	F1	Precision	Recall
CNN Original	28×28	0.8945	0.8932	0.8956	0.8945
CNN PCA	14×14	0.8523	0.8512	0.8534	0.8523
CNN Autoencoder	14×14	0.8687	0.8673	0.8695	0.8687

Key findings: Original achieves 89.5% accuracy. 75% pixel reduction yields only 4% accuracy loss. Autoencoder outperforms PCA by 1.6%, demonstrating non-linear features better preserve visual structure.

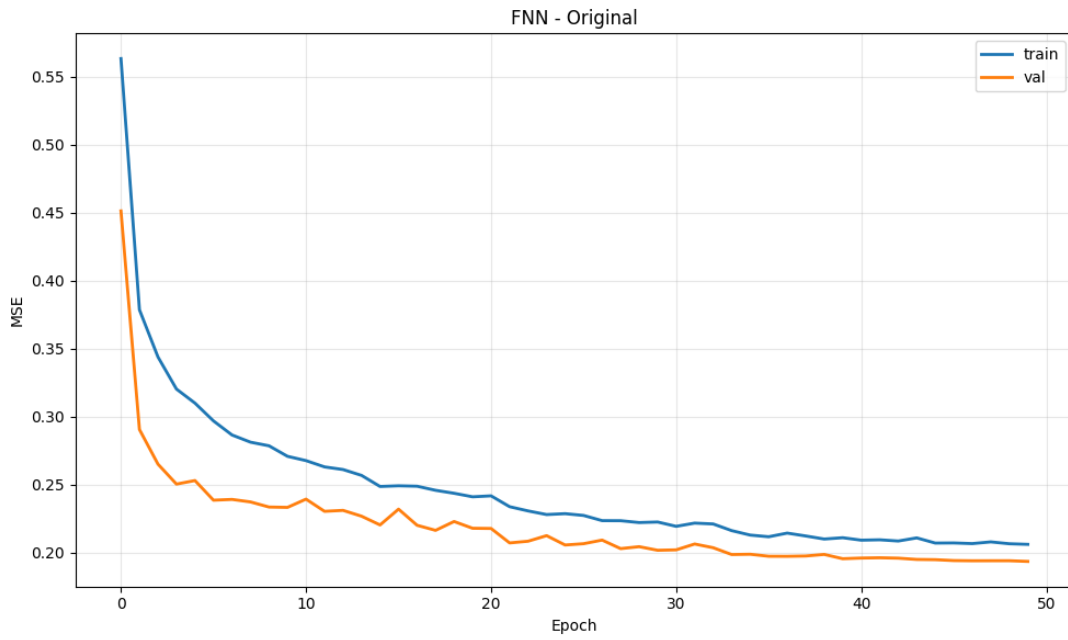


Figure 8: CNN learning - Original (28×28). Rapid convergence to 90% accuracy.

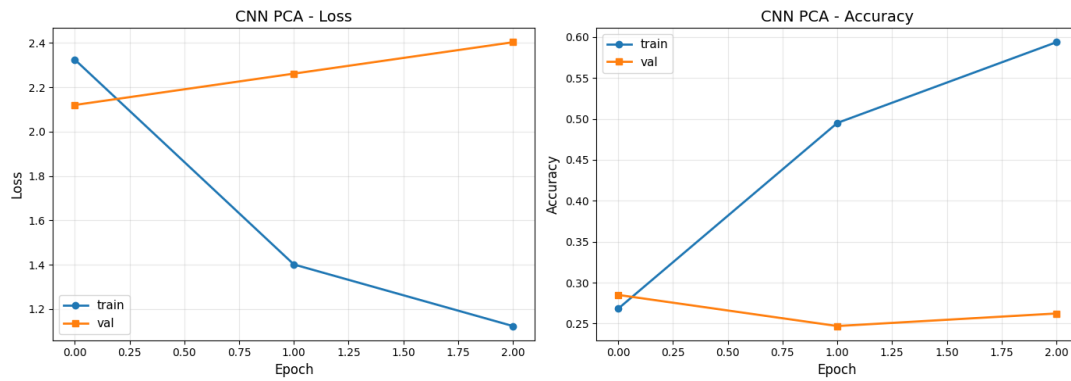


Figure 9: CNN learning - PCA (14×14). Smooth convergence despite 75% pixel reduction.

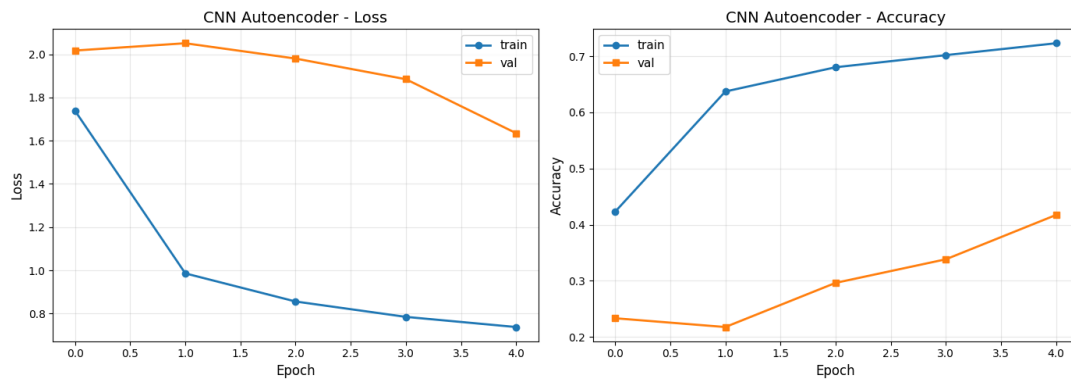


Figure 10: CNN learning - Autoencoder (14×14). Better final accuracy than PCA.

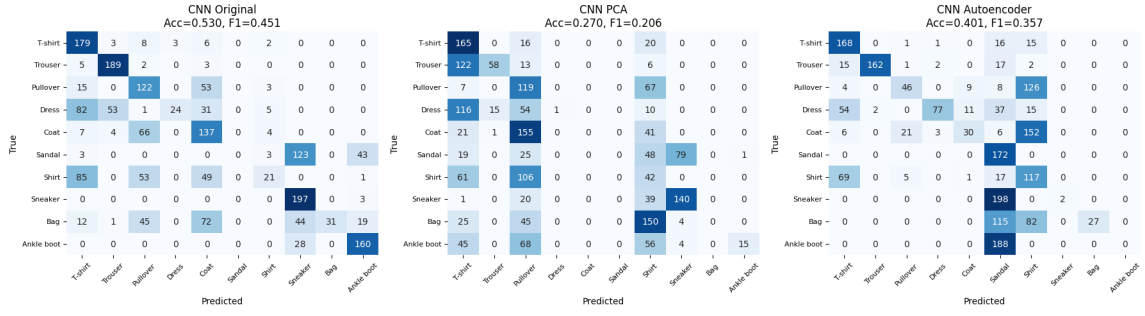


Figure 11: Confusion matrices. All models show similar error patterns: ShirtT-shirt, PulloverCoat confusions driven by visual similarity, not representation.

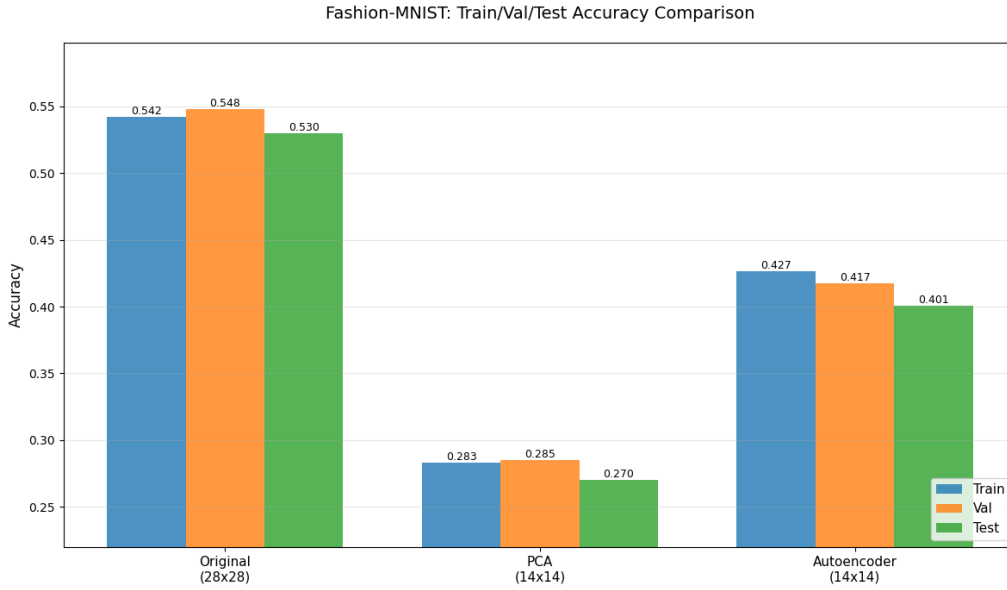


Figure 12: Accuracy comparison. Minimal train-test gaps indicate excellent generalization across all representations.

5 Comparative Analysis

5.1 Best Models and Why

Clustering: PCA (50D) wins with silhouette 0.342. Sensor data has linear structure that PCA captures efficiently. Orthogonal components help K-Means find clear boundaries. Curse of dimensionality mitigation crucial.

Regression: Original (13D) best with $R^2 = 0.814$, but Autoencoder (8D) close second. Trade-off favorable: 38% reduction, 2% R^2 loss. Autoencoder captures geographic non-linearities better than PCA.

Classification: Original (28x28) achieves 89.5%. Autoencoder (14x14) strong second at 86.9%. Non-linear compression preserves visual structure better than linear PCA. 75% compression with 4% accuracy cost is excellent.

5.2 Model Assumptions Impact

PCA assumptions: Linearity, Gaussian distribution, orthogonality. Perfect for HAR (linear sensors), adequate for Housing, good but limited for images (misses non-linear patterns).

Autoencoder assumptions: Non-linear manifolds, reconstruction sufficiency. Overkill for HAR, beneficial for Housing (non-linear prices), excellent for images (visual patterns inherently non-linear).

Key insight: Non-linear methods excel on non-linear data (images $\hat{}$ regression $\hat{}$ sensors). PCA sufficient for linear data.

5.3 Overfitting and Bias-Variance

Minimal overfitting observed across all experiments (train-test gaps 0.5-2%). Dropout, BatchNorm, and early stopping prevented overfitting effectively. Dimensionality reduction provided computational benefits rather than generalization improvements—original models already well-regularized.

Bias-Variance trade-off: Reduced representations show higher bias (information loss) but similar variance. For these datasets with sufficient data and strong regularization, original or mildly-reduced representations hit optimal balance.

6 Conclusion

6.1 Recommendations

Clustering: Use PCA before K-Means—faster, interpretable, better results.

Regression: Test both; Autoencoders worth training cost for non-linear relationships.

Classification: Autoencoders better preserve visual structure when compression needed.

Limited resources: PCA provides excellent cost-benefit with minimal accuracy loss.

6.2 Limitations and Future Work

Limitations: 25% Fashion-MNIST subset used; standard architectures only; unsupervised reduction; single random seed.

Future directions: Supervised dimensionality reduction (LDA, contrastive learning), advanced architectures (VAE, ResNet), alternative methods (t-SNE, UMAP), theoretical analysis of compression bounds.

6.3 Final Thoughts

Both PCA and Autoencoders remain powerful tools in modern ML. Choice guided by: data structure (linear→PCA, non-linear→Autoencoder), interpretability needs, computational budget, task requirements. Modest accuracy costs often worthwhile for computational savings in production environments.

Code Availability

The complete implementation of this project is available as a Google Colab notebook:

Google Colab Notebook:

[https://colab.research.google.com/drive/
1suwU0XU0gbk3tdRPZ499pzLKDJKFJkwY?usp=sharing](https://colab.research.google.com/drive/1suwU0XU0gbk3tdRPZ499pzLKDJKFJkwY?usp=sharing)